

Sequences, Progressions & Recurrences - Assimilation Book

For a learner who can often compute terms or remember AP/GP formulas, but does not yet reliably choose the cheapest representation.

The one router for this book

Before calculating, ask:

text

TERM OR SUM?

| EXPLICIT OR RECURRENT?

| LOCAL RELATION OR GLOBAL FORMULA?

| CAN NEARBY RELATIONS BE SUBTRACTED?

| CAN THE EXPRESSION TELESCOPE?

| ONLY THEN COMPUTE TERMS

The goal is not to avoid computation. The goal is to compute after structure has reduced the work.

1. RECONNECT - name the object before the formula

A sequence is an ordered list:

$a_1, a_2, a_3, \dots$

Here a_n means the term at position n .

A partial sum is a different object:

$S_n = a_1 + a_2 + \dots + a_n$.

Reconnect check

Without solving fully, write the first question you would ask.

1. 3,7,11,15,...

2. 2,6,18,54,...

3. $S_n = n^2 + 2n$; find a_n .

4. $a_{n+2} = 4a_{n+1} - 3a_n$; find a high-index term.

5. six-term moving totals keep increasing.

Good first questions are:

- difference or ratio?
- term or sum?
- explicit or recursive?
- can nearby indices be subtracted?

Term versus sum

If

$S_n = a_1 + \dots + a_n$,

then neighboring accumulations differ by exactly one contribution:

$a_n = S_n - S_{n-1}$ for $n \geq 2$.

Also $a_1 = S_1$.

Example:

$S_n = n^2 + 2n$.

Then

$a_n = S_n - S_{n-1}$

$= n^2 + 2n - [(n-1)^2 + 2(n-1)]$

$= 2n + 1$.

The important move is not expansion. It is recognizing that adjacent totals isolate one term.

2. DISCOVER - AP and GP are local invariants

The familiar formulas should come after the invariant.

Arithmetic progression

If

$$a_{n+1} - a_n = d$$

for every relevant n , the first difference is constant.

Then:

$$a_n = a_1 + (n-1)d.$$

Why $n-1$? Moving from position 1 to position n requires exactly $n-1$ equal jumps.

Geometric progression

If terms are nonzero and

$$a_{n+1}/a_n = r$$

is constant, the ratio is invariant.

Then:

$$a_n = a_1 r^{(n-1)}.$$

Close contrast

- 2,5,8,11,... -> constant difference -> AP.

- 2,6,18,54,... -> constant ratio -> GP.

- 2,5,10,17,... -> neither. Its differences 3,5,7,... have a simpler pattern.

Do not force a chapter label onto a sequence whose invariant is different.

3. MAKE SENSE - explicit and recursive are different representations

Explicit definition

$$a_n = 2n^2 - 1.$$

The requested term can be obtained directly from n .

Recursive definition

$$a_{n+1} = a_n + 2n + 1, \text{ together with } a_1 = 1.$$

The next term depends on earlier information.

A recurrence by itself is usually not a complete sequence. It needs:

- the index range where the rule applies;
- enough initial values.

A second-order recurrence such as

$$a_{n+2} = 3a_{n+1} - 2a_n$$

normally needs two initial terms.

How to verify a proposed explicit formula

Suppose a recurrence is

$$a_{n+2} = 3a_{n+1} - 2a_n,$$

with $a_1 = 1$, $a_2 = 4$.

Someone proposes:

$$a_n = 3 \cdot 2^{(n-1)} - 2.$$

Verification has two parts.

1. Initial values

- at $n=1$: $3 \cdot 2 = 6$;
- at $n=2$: $6 \cdot 2 = 12$.

2. Recurrence for every allowed n

Substitute the formula into both sides and simplify. Matching a few early terms is evidence, not a proof for all indices.

Adopt: a recurrence means dependency + range + initialization.

4. DISCOVER - subtract nearby windows

Let a fixed k -term window be

$$W_i = a_i + a_{i+1} + \dots + a_{i+k-1}.$$

The next window is

$$W_{i+1} = a_{i+1} + \dots + a_{i+k}.$$

Subtract:

$$W_{i+1} - W_i = a_{i+k} - a_i.$$

The $k-1$ shared terms disappear.

If the windows are averages, dividing by the same positive k does not change which average is larger. Multiply by k first, then subtract the sums.

Historical mechanism

For the validated 2025 Q26 anchor:

- increasing 4-term averages give $a_{i+4} > a_i$;
- decreasing 7-term averages give $a_{i+7} < a_i$.

The original surface is "averages." The useful representation is an index-shift inequality graph.

Adopt: a moving-window problem is usually a subtraction problem before it is an averaging problem.

5. MAKE SENSE - transform a recurrence before iterating

Consider

$$a_{n+2} = 4a_{n+1} - 3a_n.$$

Subtract a_{n+1} from both sides:

$$a_{n+2} - a_{n+1} = 3(a_{n+1} - a_n).$$

Define the first-difference sequence

$$d_n = a_{n+1} - a_n.$$

Then:

$$d_{n+1} = 3d_n.$$

The original second-order recurrence has become a first-order GP for differences.

This is the local/global decision:

- global brute force: compute many a_n ;
- local structural route: find a simpler quantity that evolves predictably.

The transform is not always "first differences." It may be:

- a sum/difference of nearby terms;
- a ratio;
- an alternating transform;
- a neighboring-term invariant.

6. MAKE SENSE - a high-index neighboring-term invariant

A second-order recurrence can preserve a simple determinant-like combination.

Suppose

$$a_{n+2} = p a_{n+1} + q a_n.$$

Define

$$D_n = a_n^2 - a_{n-1} a_{n+1}.$$

Now compare neighboring indices:

$$D_{n+1} = a_{n+1}^2 - a_n a_{n+2}$$

Use the recurrence:

$$= a_{n+1}^2 - a_n (p a_{n+1} + q a_n)$$

Group the first two terms:

$$= a_{n+1} (a_{n+1} - p a_n) - q a_n^2.$$

From the recurrence one index earlier,

$$a_{n+1} - p a_n = q a_{n-1}.$$

Therefore

$$D_{n+1} = q a_{n-1} a_{n+1} - q a_n^2 \\ = -q(a_n^2 - a_{n-1} a_{n+1})$$

so

$$D_{n+1} = -q D_n.$$

A complicated expression involving three high-index terms has become a one-step GP.

Anchor connection

For the validated 2023 Q10 recurrence

$$a_{n+2} = -4a_{n+1} - 7a_n,$$

we have $q = -7$, hence

$$D_{n+1} = 7D_n.$$

That is why computing a_{49}, a_{50}, a_{51} directly is the wrong representation.

Author-created check

$$\text{Let } b_0 = 0, b_1 = 1 \text{ and} \\ b_{n+2} = -4b_{n+1} - 7b_n.$$

Then $D_1 = 1$, so

$$\mathbf{D_{20} = 7^{19}.}$$

Stop at this recurrence-invariant endpoint in this chapter. Any later question about the number of divisors of 7^{19} belongs to the subsequent number-theory treatment of prime powers and divisor counts.

The raw terms become huge; the invariant stays tiny.

7. DISCOVER - telescoping is cancellation across a sum

Consider

$$1/[k(k+1)].$$

Split it:

$$1/[k(k+1)] = 1/k - 1/(k+1).$$

Then

$$\sum_{k=1}^n 1/[k(k+1)]$$

becomes

$$(1 - 1/2) + (1/2 - 1/3) + \dots + (1/n - 1/(n+1)).$$

Every internal term cancels:

$$= 1 - 1/(n+1) = n/(n+1).$$

Recognition cue

Neighboring factors such as $k(k+1)$ or $(k-1)k$ are a clue, not a guarantee. You must verify an exact decomposition $F(k) - F(k+1)$.

Same idea, different scale

- $S_n - S_{n-1}$ cancels an accumulation.
- $W_{i+1} - W_i$ cancels overlapping windows.
- a shifted recurrence cancels repeated structure.
- telescoping cancels internal summands.

All are local comparisons between neighboring indices.

8. TRY - attempt first, then optional help

For each problem, attempt a first line before reading any prompt.

Try A - term from accumulation

$$S_n = 3n^2 - n. \text{ Find } a_n.$$

If stuck:

- Recognition prompt: one term is hidden between two neighboring totals.
- Structural prompt: compare S_n and S_{n-1} .
- Full support: write $a_n = S_n - S_{n-1}$.

Try B - moving window

Every 6-term sum is larger than the preceding 6-term sum. State the direct term inequality.

If stuck:

- Recognition prompt: the two sums share five terms.
- Structural prompt: subtract adjacent 6-term windows.
- Full support: write $W_{i+1} - W_i = a_{i+6} - a_i$.

Try C - recurrence transform

$$a_{n+2} = 5a_{n+1} - 4a_n.$$

Find a recurrence for $d_n = a_{n+1} - a_n$.

If stuck:

- Recognition prompt: compare neighboring terms rather than raw values.
- Structural prompt: rewrite in first differences.
- Full support: $a_{n+2} - a_{n+1} = 4(a_{n+1} - a_n)$.

9. DIAGNOSE - why the tempting start fails

| Wrong move | Why it is tempting | Missing link | Repair |

|---|---|---|---|

set $a_n = S_n$	both use index n	representation	term = adjacent-sum difference
call every regular list AP/GP	formulas are familiar	recognition	test difference/ratio invariant
ignore initial values	recurrence looks like a formula	semantics	recurrence + range + initialization
verify only first few terms	examples match	proof/check	initials + recurrence for all valid n
compute 50 recurrence terms	procedure is obvious	method selection	search local transform/invariant
expand two moving averages	surface says "average"	representation	cancel overlapping windows
telescope any rational sum	one example was memorable	boundary	prove exact neighbor difference
derive a counting recurrence here	notation looks identical	ownership/model	the counting chapter defines the state first
lose one index	nearby subscripts look similar	execution	write both shifted relations explicitly

10. FADE - support decreases across the set

The available support now fades.

Full support available

$$S_n = 2n^2 + 3n. \text{ Find } a_n.$$

Maximum support if needed:

$$a_n = S_n - S_{n-1}.$$

Structural prompt only

$$a_{n+2} = 4a_{n+1} - 3a_n. \text{ Find a simpler sequence.}$$

Maximum support:

study first differences.

Recognition prompt only

Evaluate

$$\sum_{k=2}^n \frac{1}{k(k-1)}.$$

Maximum support:

look for neighboring-factor cancellation.

Independent

A sequence has all 4-term sums equal. Prove a periodicity statement.

No hint.

11. ADOPT - the first-move rules

1. Term or sum? Name the object.
2. Explicit or recurrent? Decide whether direct substitution or dependency is present.
3. Local or global? Prefer the representation that exposes the target.
4. Nearby relations? Subtract before iterating.
5. Telescoping? Seek an exact neighbor difference.
6. High index? Search transformed sequence/invariant.

7. Counting state? This chapter supplies recurrence notation only; a later combinatorics chapter must define the state and derive a counting recurrence.

Say this in your own words:

"When neighboring indices share most of their structure, I compare them before I calculate."

12. TRANSFER - changed surfaces

Rolling measurements

A sensor records daily readings $r_1, r_2, \dots$. Every 6-day total is larger than the previous 6-day total.

The first useful line is not an average formula. It is:

$$r_{i+6} > r_i.$$

Audit quantity for machine readings

Readings satisfy

$$x_{n+2} = 2x_{n+1} + 3x_n.$$

Define

$$Q_n = x_n^2 - x_{n-1}x_{n+1}.$$

The same derivation gives:

$$Q_{n+1} = -3Q_n.$$

Different coefficients, same invariant mechanism.

A recurrence from tilings

Suppose a later tiling chapter states a recurrence $t_n = t_{n-1} + t_{n-2}$.

From this chapter you can:

- read the notation;
- use initial values;
- verify or manipulate the recurrence.

But the statement "why is the count equal to $t_{n-1} + t_{n-2}$?" belongs to the counting-model chapter. It requires a state and a disjoint, exhaustive first-step partition.

13. Historical anchors

- IOQM 2025 Q26 - window comparison and high-index cancellation by local inequalities; verified answer 10.
- IOQM 2023 Q10 - neighboring-term determinant invariant under a second-order recurrence; verified answer 51.

Historical wording remains controlled by the validated papers. The exercises in this book that are not explicitly identified by year/question are author-created.

14. Exit test

You own this topic when you can answer all six questions for each major mechanism:

1. What did I notice?
2. Why does the method work?
3. What clue triggers it?
4. What similar-looking problem needs a different start?
5. Can I write the first two useful lines without help?
6. Can I solve a changed-surface version?

Formula recall alone is not the exit condition.

ALG-04 - First-Step Reference

This is a compression layer. Use it after the Assimilation Book.

Master router

text

TERM OR SUM?

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LOCAL RELATION OR GLOBAL FORMULA?

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CAN NEARBY RELATIONS BE SUBTRACTED?

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CAN THE EXPRESSION TELESCOPE?

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ONLY THEN COMPUTE TERMS

Recognition atlas

| Visible structure | Ask | First useful line |

|---|---|---|

| S_n given, a_n asked | accumulation or term? | $a_n = S_n - S_{n-1}$ |

| constant local change | difference or ratio? | compute $a_{n+1} - a_n$ |

| repeated multiplication | ratio constant? | compute a_{n+1}/a_n |

| $a_n = f(n)$ | explicit? | substitute requested n |

| earlier terms on RHS | recurrent? | write initials + valid index range |

| moving sums/averages | overlapping windows? | subtract adjacent windows |

| high-index recurrence | can a transform simplify? | write nearby-index copies |

| $k(k+1) / (k-1)k$ | exact telescope? | seek $F(k) - F(k+1)$ |

| squares/products of neighboring recurrence terms | invariant candidate? | compare candidate at $n, n+1$ |

| tilings/paths/states | where did recurrence come from? | route state derivation to the later combinatorics chapter |

AP / GP card

AP:

$a_{n+1} - a_n = d$.

GP:

$a_{n+1}/a_n = r$ for nonzero terms.

Do not decide from appearance. Test the invariant.

Recurrence interface card

Notation

$a_{n+2} = F(a_{n+1}, a_n)$ relates indexed terms.

Semantics

A recurrence is valid on a stated index range.

Initialization

An order-2 recurrence normally needs two starting values.

Explicit versus recursive

- explicit: $a_n = f(n)$;

- recursive: a_n depends on earlier terms.

Verification

1. check all initial values;
2. substitute the candidate into the recurrence;
3. prove the equality for every allowed index.

Local cancellation

Before iterating:

- write the relation at neighboring indices;
- subtract/combine;
- test a transformed sequence.

Window card

For

$W_i = a_i + \dots + a_{i+k-1}$,

$W_{i+1} - W_i = a_{i+k} - a_i$.

Fixed-size averages have the same ordering as their sums.

Telescoping card

Target form:

$$u_k = F(k) - F(k+1).$$

Then the sum collapses to boundary terms.

Neighboring-term invariant card

For

$$a_{n+2} = p a_{n+1} + q a_n,$$

try

$$D_n = a_n^2 - a_{n-1} a_{n+1}.$$

Then

$$D_{n+1} = -q D_n.$$

Use only after deriving/checking it for the given recurrence.

Contrast strip

- term vs partial sum;
- AP vs GP;
- explicit vs recursive;
- local relation vs global formula;
- direct iteration vs shifted-relation subtraction;
- ordinary rational sum vs true telescoping sum;
- raw high-index terms vs invariant;
- algebraic recurrence vs counting-state recurrence;
- deterministic state evolution vs adversarial game;
- checking examples vs verifying a recurrence.

Fast checks

Before finalizing:

- Did I preserve the correct index?
- Did I state enough initial values?
- Did I divide by a quantity that could be zero?
- Did I prove the telescope decomposition?
- Did I use the recurrence only where it is valid?
- If the recurrence counts objects, was the state decomposition justified elsewhere?

ALG-04 - Recognition and First-Line Lab

Make an independent first attempt. Write only the recognition statement or first useful mathematical line; do not finish the whole problem unless needed to justify the first move.

Core recognition

1. $S_n = 2n^2 + 3n$; asked for a_n .
2. **5,9,13,17,....**
3. **3,12,48,192,....**
4. **2,5,10,17,26,....**
5. $a_n = 4n^2 - 1$; asked for a_{100} .
6. $a_{n+1} = a_n + 2n + 1$, $a_1 = 1$; asked what information defines the sequence.
7. 5-term moving sums are strictly increasing.
8. 7-term moving averages are strictly decreasing.
9. $a_{n+2} = 5a_{n+1} - 4a_n$; look for a recurrence for first differences.
10. $\sum 1/[k(k+1)]$.
11. $\sum 1/(k^2 + 1)$.
12. $b_{n+2} = -4b_{n+1} - 7b_n$; target contains $b_n^2 - b_{n-1}b_{n+1}$.

Close contrasts

13. A: 2,6,10,14,...

B: 2,6,18,54,...

State the local invariant that separates them.

14. A: $a_n = 3n + 1$

B: $a_{n+1} = a_n + 3$, $a_1 = 4$

Which is explicit and which is recursive?

15. A: all 4-term sums are equal.

B: S_n is known explicitly.

Which nearby subtraction should be written in each case?

16. A: $1/[k(k+1)]$.

B: $1/(k^2 + 1)$.

Which one has an immediate neighboring-factor telescope?

17. A recurrence is supplied algebraically with two initial values.

A recurrence is claimed to count tilings.

What must be justified in the second problem that is not needed in the first?

18. A deterministic process has fixed transitions.

A two-player game has strategic choices.

Which neighboring topic owns the second structure?

Verification recognition

19. A proposed formula matches the first five terms of a recurrence. What is still missing for verification?

20. A second-order recurrence is written with no initial values. What information is missing before it defines a unique sequence?

ALG-04 ? Practice and Transfer Bank

First attempt every item independently. The first section repairs foundations; later sections increase decision complexity.

F0 ? Foundation

1. Find the 15th term of 4,7,10,....
2. Find the 8th term of 3,6,12,....
3. If $S_n = n(n+1)$, find a_n .
4. Classify each representation:
- $a_n = 2n^2 - 1$;
- $a_{n+1} = a_n + 2n + 1$, $a_1 = 1$.
State what initialization the recursive form needs.

F1 ? Direct structure

5. Determine whether 2,5,10,17,26,... is AP, GP, or neither. Identify a simpler pattern in its first differences.
6. If every 4-term sum is larger than the preceding 4-term sum, derive a direct inequality involving a_i and a_{i+4} .
7. Evaluate $\sum_{k=1}^{20} \frac{1}{k(k+1)}$.
8. Verify, not merely test, that $a_n = 3 \cdot 2^{(n-1)} - 2$ satisfies
 $a_{n+2} = 3a_{n+1} - 2a_n$,
 $a_1 = 1, a_2 = 4$.

F2 ? Standard

9. If $S_n = 2n^2 - n$, show that the resulting sequence is an AP and find its common difference.
10. Given $a_{n+2} = 4a_{n+1} - 3a_n$, $a_1 = 2, a_2 = 5$, transform to first differences and find a_8 .
11. Evaluate $\sum_{k=2}^n \frac{1}{k(k-1)}$.
12. All 3-term sums of a sequence are equal. Prove the sequence is periodic with period dividing 3.

F3 ? Disguised

13. Five-term averages are increasing and seven-term averages are decreasing. Translate both statements into term inequalities before doing anything else.
14. Let $b_0 = 0, b_1 = 1$ and $b_{n+2} = -4b_{n+1} - 7b_n$.
Define $D_n = b_n^2 - b_{n-1}b_{n+1}$.
Find D_{20} without computing b_{20} . Stop once the invariant has determined D_{20} ; no divisor-count formula is needed here.
15. Let $T_n = a_1 + \dots + a_n$ and $T_n = 3T_{n-1} + 2$ for $n \geq 2$.
Rewrite a_n directly in terms of T_{n-1} .
16. Evaluate $\sum_{k=1}^n \frac{1}{[(2k-1)(2k+1)]}$ by telescoping.

F4 ? Preliminary-style

17. A sequence satisfies $a_{n+2} - a_{n+1} = 2(a_{n+1} - a_n)$,
with $a_1 = 2, a_2 = 5$.
Find a_{10} efficiently.
18. All 4-term sums are equal. If $a_1 = 1, a_2 = 2, a_3 = 4, a_4 = 8$,
find a_{99} .
19. If $S_n = n^3$, find a_n . Explain why n^3 is not a valid discrete argument even though the resulting expression is also quadratic.
20. A student says: "The first three ratios are equal, so the infinite sequence is definitely a GP." Give the smallest logical objection.

Support-fading tracks

These four tasks deliberately reduce the maximum available support. The support labels below are learner-facing descriptions rather than internal control codes.

Full support available

21. $S_n = 5n^2 + n$. Find a_n .
- First prompt if needed: compare neighboring accumulated totals.
- Second prompt if needed: use $S_n - S_{n-1}$.
- Execution prompt if needed: start with $a_n = (5n^2 + n) - [5(n-1)^2 + (n-1)]$.

Medium support available

22. $a_{n+2} = 6a_{n+1} - 5a_n$. Find a simpler recurrence.
- First prompt if needed: compare neighboring terms.
- Second prompt if needed: study first differences.

Light support available

23. Evaluate $\sum_{k=3}^n \frac{1}{[(k-1)k]}$.

- Prompt if needed: consecutive factors suggest local cancellation.

Independent

24. Every 5-term sum equals the preceding 5-term sum. Prove the strongest simple periodicity statement you can.

Transfer bank

25. Context change ? rolling totals.

A sensor's consecutive 6-day totals are strictly increasing. Without computing any total explicitly, prove a direct inequality between two readings six days apart.

26. Representation change ? accumulation to contribution.

A layered design has cumulative cost $C_n = n(n+1)/2$ after n layers. Find the cost of layer n using no summation formula.

27. Context change ? invariant audit.

Machine readings satisfy

$$x_{n+2} = 2x_{n+1} + 3x_n.$$

For $Q_n = x_n^2 - x_{n-1}x_{n+1}$, determine Q_{n+1}/Q_n whenever $Q_n \neq 0$.

28. Representation change ? telescope recognition.

Compare

$$\sum (1/k - 1/(k+1))$$

with

$$\sum 1/[k(k+1)].$$

Explain why they are the same cancellation written in different representations.

29. Close boundary ? local versus global.

Problem A gives an explicit $a_n = f(n)$ and asks for a_{500} .

Problem B gives a second-order recurrence and asks for a_{500} .

Explain why ?high index? alone does not determine the method.

30. Ownership bridge ? counting state.

A tiling count is asserted to satisfy $t_n = t_{n-1} + t_{n-2}$.

List what sequence algebra can verify once this recurrence is supplied, and what must be proved from the counting model before the recurrence is legitimate.

ALG-04 ? Independent Mixed Mastery Check

No default hints and no method labels. First attempt is independent.

Part A ? Notice and first line

1. The partial sums are $S_n = 4n^2 + n$. Write the first useful line for finding a_n .
2. Every 6-term sum is greater than the preceding 6-term sum. Write the first useful line.
3. $a_n = 2^n + 3$. The target is a_{50} . State why no recurrence iteration is needed.
4. $a_{n+2} = 5a_{n+1} - 4a_n$. The target is a high-index term. Write a useful transformed relation before computing terms.

Part B ? Full mixed solve

5. The partial sums are $S_n = 4n^2 + n$. Find a_n .
6. $a_1 = 1, a_2 = 4$ and $a_{n+2} = 5a_{n+1} - 4a_n$. Find a_8 efficiently.
7. Evaluate $\sum_{k=1}^{50} \frac{1}{k(k+1)}$.
8. Every 4-term sum is equal to the preceding one. If $a_1 = 1, a_2 = 2, a_3 = 4, a_4 = 8$, find a_{99} .
9. Let $b_0 = 0, b_1 = 1$,
 $b_{n+2} = -4b_{n+1} - 7b_n$,
and $D_n = b_n^2 - b_{n-1}b_{n+1}$.
Find D_{20} without computing b_{20} . Stop at the invariant result; divisor-count doctrine is outside this topic.

Part C ? Same surface, different decision

10. Sequence A: 2, 6, 10, 14,
Sequence B: 2, 6, 18, 54,
State the invariant that classifies each.
11. Representation A: $a_n = 3n + 1$.
Representation B: $a_{n+1} = a_n + 3, a_1 = 4$.
Identify explicit versus recursive form and state why the initial value is necessary in B.
12. Compare:
- $\sum_{k=1}^n \frac{1}{k(k+1)}$;
- $\sum_{k=1}^n \frac{1}{(k^2 + 1)}$.
Explain why the first has an immediate neighboring-term telescope but the second should not be declared telescoping from appearance alone.

Part D ? Changed-surface transfer

13. A shop records daily sales $r_1, r_2, \dots$. Every 5-day rolling total is larger than the preceding 5-day total.
Prove the corresponding term inequality.
14. Machine readings satisfy
 $x_{n+2} = 2x_{n+1} + 3x_n$.
Define
 $Q_n = x_n^2 - x_{n-1}x_{n+1}$.
Derive the recurrence for Q_n .

Part E ? WHY-NOT and verification

15. A proposed explicit formula agrees with the first five terms generated by a recurrence. Explain why this alone is not a proof, and state the complete verification test.
16. A student sees a tiling count with the recurrence $t_n = t_{n-1} + t_{n-2}$ and says, "It is Fibonacci, so the recurrence is proved." Explain precisely what must still be established before the recurrence is valid for the counting problem.

Ownership check

For any two problems above, state:

- the visible clue;
- the hidden structure;
- the first useful line;
- one tempting but inferior or invalid nearby method.